TWINX: Digital Twin for Robot Algorithm Testing

Suitable robot: Unitree Go2 or Unitree G1

Create a digital twin environment that represents both the physical robot and the laboratory or university space, enabling navigation, perception, and interaction algorithms to be developed and tested safely in simulation before deployment on the real robot. The project should include a realistic robot simulation model, a virtual representation of the selected environment, and a connection to ROS 2 so that the same or closely related software components can be used in both simulation and real-world experiments.

You should test selected navigation or interaction scenarios in simulation, analyse robot behaviour under different conditions, and then transfer the developed algorithms to the physical platform. The project should investigate the differences between simulated and real-world performance, including issues such as sensor noise, localisation accuracy, obstacle detection, motion behaviour, and environmental variability.

The final work should include simulation-to-real transfer experiments, quantitative comparison of performance in both environments, and a validation report identifying limitations, transfer challenges, and possible improvements to reduce the simulation-to-reality gap.

PROJECT AIM

Develop a digital-twin test environment in which robot navigation, perception or interaction algorithms can be tested safely before they are deployed. The research question is how closely the simulation predicts real robot behaviour and where the simulation-to-reality gap appears.

POSSIBLE RESEARCH QUESTIONS

- How accurately does the digital twin predict the physical robot's behaviour for the selected navigation, perception, or interaction task?
- Which simulation parameters or calibration steps most effectively reduce the observed simulation-to-reality gap?

WHAT YOU WILL BUILD

- Model the selected robot and a limited laboratory/university environment.
- Connect the twin to ROS 2 so software modules can be reused between simulation and reality.
- Design repeatable scenarios and log comparable metrics in both environments.
- Analyse localisation, sensing, motion and environmental differences.

START HERE

1. Select one primary use case (for example navigation) rather than trying to twin every robot capability.
2. Define the physical environment and the measurable variables before modelling it.

3. Create one benchmark scenario that can later be repeated exactly on the real robot.

DEVELOPMENT PATH

The digital twin is the first deliverable, not a temporary placeholder. After the robot arrives, use the same test scripts and metrics to quantify transfer quality and identify which simulation assumptions need calibration.

EXPECTED DELIVERABLES

- Reusable digital-twin environment and ROS 2 interface.
- Simulation/real benchmark suite with quantitative comparison.
- Validation report, calibrated model, code and thesis.